

INTERNSHIP PROJECT REPORT

Superstore Sales Analysis Dashboard using Power BI

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1. Introduction

As part of my internship at [Company Name], I was assigned a project to design and build an interactive **Superstore Sales Analysis Dashboard** using **Microsoft Power BI**. Retail and superstore businesses generate large volumes of transactional data every day, and turning this data into clear visual insights is essential for tracking performance and guiding business decisions. Power BI is one of the most widely used business intelligence tools for transforming raw sales data into interactive, easy-to-understand dashboards.

This project focuses on analyzing sales and profit data of a superstore across product categories, sub-categories, regions, customer segments, and time periods, and presenting the results through a multi-page interactive dashboard. The dashboard allows stakeholders to view overall sales performance at a glance and drill down into specific areas such as region-wise sales, product-wise performance, top customers, and monthly sales trends.

2. Objectives of the Project

- To understand the fundamentals of data visualization and business intelligence using Power BI.
- To clean, transform, and model raw superstore sales data for analysis.
- To design an interactive dashboard summarizing total sales, profit, and order count.
- To analyze sales performance across product categories, sub-categories, and regions.
- To identify the top customers contributing to overall sales.
- To study sales trends over different months and by customer segment for better business planning.
- To present actionable insights that support data-driven decision-making.

3. Tools and Technologies Used

Tool / Technology	Purpose
Microsoft Power BI Desktop	Data modeling, visualization, and dashboard creation
Power Query Editor	Data cleaning and transformation
DAX (Data Analysis Expressions)	Creating calculated columns and measures
Excel / CSV (Superstore dataset)	Source data format for the sales dataset
Power BI Service (optional)	Publishing and sharing the dashboard online

4. About the Dataset

The dataset used for this project is the well-known **Superstore sales dataset**, which contains order-level transactional records for a retail superstore. Based on the dashboard built, the key fields in the dataset include:

- **Sales** – Revenue generated from each order.
- **Profit** – Net profit earned on each order.
- **Row ID / Order Count** – Unique identifier used to count the number of orders.
- **Region** – Geographic region of sale (East, West, Central, South).
- **State or Province** – Location of the customer, shown on a map visual.
- **Product Category** – High-level grouping (Technology, Furniture, Office Supplies).
- **Product Sub-Category** – Detailed grouping (Office Machines, Chairs & Chairmats, Telephones and Communication, Tables, Binders, Storage, Bookcases, Copiers and Fax, Office Furnishings, Computer Peripherals, Appliances, Paper, Pens & Art Supplies, Envelopes, Scissors & Rulers, Labels, Rubber Bands).
- **Customer Name** – Name of the customer placing the order.
- **Customer Segment** – Consumer, Corporate, Home Office, Small Business.
- **Order Date (Day / Month)** – Time period of the order, covering January to June.

The dataset was cleaned and transformed in Power Query to handle missing values, ensure correct data types (dates, numbers, and text), and create relationships between tables where required, before being loaded into the Power BI data model for analysis.

5. Methodology

- 1 **Data Collection:** Gathered the raw Superstore sales data in tabular format (Excel/CSV).
- 2 **Data Cleaning:** Used Power Query Editor to remove duplicates, fix data types, and handle missing or inconsistent values.
- 3 **Data Modeling:** Organized the data into a structured model and created relationships between relevant tables (e.g., Orders, Products, Customers, Region).
- 4 **DAX Measures:** Created calculated measures such as Sum of Sales, Sum of Profit, and Count of Row ID for use across visuals.
- 5 **Dashboard Design:** Designed a multi-page report with a navigation panel (Dashboard / Details buttons), KPI cards, bar charts, a donut chart, a funnel chart, area/line charts, a gauge, a ribbon chart, and a map visual.
- 6 **Interactivity:** Added navigation buttons and cross-filtering so that visuals respond consistently as the user explores the report.
- 7 **Testing & Validation:** Verified that totals and trends in the visuals matched the underlying data before finalizing the report.

6. Dashboard Design and Visualizations

The final Power BI report is organized as a navigable dashboard with a side panel (Dashboard / Details) and multiple sections combining summary KPIs with detailed breakdowns by category, region, customer, and time period.

6.1 Dashboard Overview

The main dashboard page provides a high-level summary of the superstore's performance using three KPI cards (Sum of Sales, Sum of Profit, and Count of Row ID / total orders), along with a bar chart of Sales by Product Sub-Category, a donut chart of Sales by Product Category, an area chart of Sales by Day and Month, and a table listing top customers by Sum of Sales.

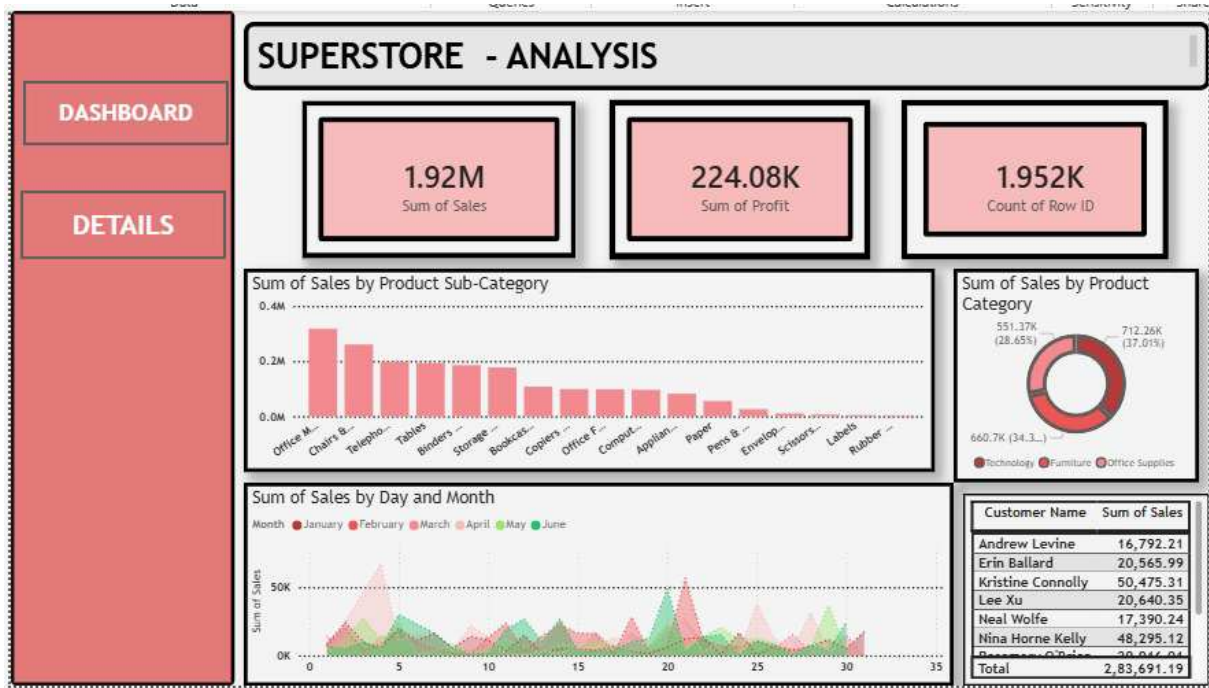


Figure 1: Superstore Analysis Dashboard – Main overview page

6.2 Regional and Category-wise Sales Analysis

This section drills into geographic and product performance. A bar chart shows Sales by Region (East leading, followed by West, Central, and South), a pie chart shows Sales by Product Category split across Technology, Furniture, and Office Supplies, a funnel chart ranks Sales by Product Sub-Category from Office Machines down to Rubber Bands, a line chart tracks Sales by Month showing a clear peak in April, and a map visual plots sales concentration by State or Province across the United States.

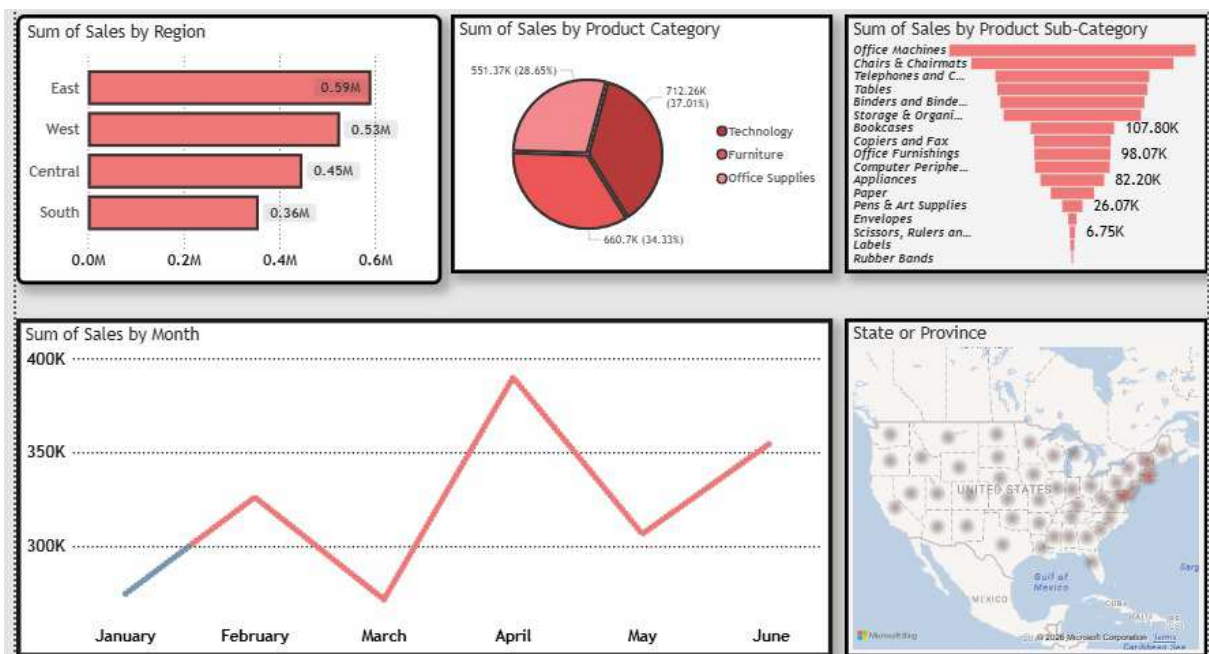


Figure 2: Regional and Category-wise Sales Analysis page

6.3 Segment and Profitability Analysis

The final section focuses on customer segments and profitability. An area chart shows Sales by Month and Customer Segment (Consumer, Corporate, Home Office, Small Business), a area/line chart tracks Sum of Profit by Day (averaging around 7.23K), a gauge visual shows Sum of Sales (1.92M) against a target range up to 3.85M, and a combo chart compares Sum of Sales (bars) and Sum of Profit (line) across the three product categories – showing that while Technology has the highest sales, Office Supplies delivers the highest profit relative to its sales.

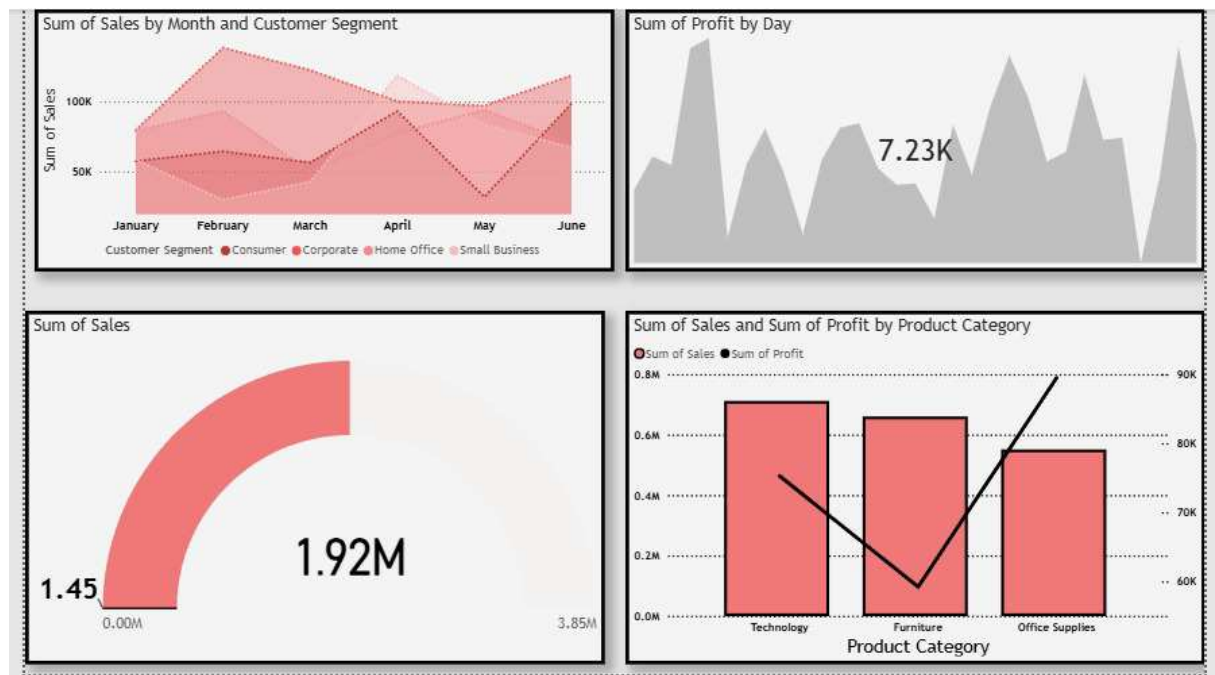


Figure 3: Segment and Profitability Analysis page

7. Key Performance Indicators (KPIs)

KPI	Value
Sum of Sales	1.92M
Sum of Profit	224.08K
Count of Row ID (Orders)	1.952K
Average Profit per Day	7.23K

Note: KPI values shown reflect the figures visible on the dashboard cards; update this table if the underlying data or filters change.

8. Findings and Insights

- **Technology is the top category by sales** – contributing about 37% of total sales, followed by Office Supplies (34%) and Furniture (29%).
- **Office Machines and Chairs & Chairmats lead sub-category sales** – both significantly outperform other sub-categories such as Labels and Rubber Bands.
- **East region generates the highest sales** – followed by West, Central, and South, indicating stronger demand or customer base in the East.
- **Sales peaked in April** – the monthly trend shows a sharp rise in April compared to a dip in March, useful for planning seasonal promotions and inventory.
- **Office Supplies is more profit-efficient** – despite lower sales than Technology, it shows a strong profit line relative to its sales in the category comparison chart, while Furniture shows comparatively lower profit.
- **Top customers vary widely in contribution** – customers like Kristine Connolly and Nina Horne Kelly contribute noticeably higher sales than others in the top customer list.
- **Customer segments show fluctuating demand** – the Consumer and Corporate segments show the most variation across months, useful for targeted marketing.

9. Challenges Faced

- Cleaning and standardizing raw transactional data before it could be used reliably in visuals.
- Choosing the right chart types (funnel, gauge, ribbon, map) to represent different business questions clearly.
- Designing DAX measures to correctly aggregate Sales and Profit across categories and time periods.
- Building a navigation panel (Dashboard / Details) while keeping the report structure simple to use.
- Arranging multiple visuals on a single page while keeping the dashboard readable and uncluttered.

10. Learning Outcomes

- Gained hands-on experience with Power BI Desktop, from data import to dashboard publishing.
- Learned how to use Power Query for data cleaning and transformation.
- Understood how to write DAX measures for aggregations and KPIs.
- Learned to use varied visual types – funnel charts, gauges, maps, and ribbon charts – to represent different kinds of business questions.
- Learned how to build report navigation using buttons and bookmarks in Power BI.
- Improved the ability to derive business insights from visualized data rather than raw numbers.
- Developed a better understanding of how retail sales and profitability data is used in real business decision-making.

11. Conclusion

This internship project provided valuable exposure to real-world data analytics and business intelligence practices using Microsoft Power BI. By building a multi-page Superstore Sales Analysis Dashboard, I was able to translate raw transactional data into clear, actionable insights covering overall performance, regional and category-wise trends, top customers, and profitability by product category. The project strengthened my skills in data cleaning, data modeling, DAX, and visual storytelling, and gave me a practical understanding of how organizations use dashboards to monitor performance and support decision-making. Overall, this internship has been a valuable learning experience that bridges academic knowledge with practical, industry-relevant skills.

12. References

- Microsoft Power BI Documentation – <https://learn.microsoft.com/power-bi/>
- DAX Function Reference – <https://learn.microsoft.com/dax/>
- Sample Superstore dataset used during the internship period.